

East West University

Project Report
CSE479 - Web Programming
Section: 1
Project Title: Progga - Online Learning Management System (LMS)

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1. Introduction

With the amplification of digital platforms in education, the demand for efficient learning management systems that can seamlessly connect educators and learners has become a necessity. Schools and universities, as the primary educational institutions, are struggling to handle the various aspects of teaching and learning. Most students do not find courses that match their level of skills and areas of interest, which leads to an inefficient way of learning and low engagement. On the other hand, educators are handling different tools for course creation, content delivery, assessment, and communication, which causes them to have more administrative work and less teaching time. Present systems are devoid of integrated solutions that can simultaneously satisfy a learner's demand for personalized course discovery and an educator's need for management capabilities. Progga is an online Learning Management System (LMS) that aims to solve these problems by offering a single platform that meets the needs of students and teachers in academic environments.

This project is a fully-fledged web-based system built with PHP, HTML, CSS, JavaScript, and MySQL that covers the educational workflow from end to end. Progga equips students with a simple and interactive course marketplace where they can inspect and filter courses according to three levels of difficulty (beginner, intermediate, and advanced) and exact topic areas, with alternatives for free and discounted courses. The student, after enrolling in a course, may view video lectures, enter discussion forums for peer and instructor interaction, take quizzes, and rate a course based on the learning experience.

The platform provides educators with efficient course administration instruments allowing them to shape and systematize courses, upload multimedia content such as video lectures, create and grade tests, communicate directly with students, and check their teaching performance through analytics dashboards that monitor course ratings and earnings. Along with this, the system has various reporting features that provide both groups of users with progress, engagement, and learning outcomes' records. This fully realized platform is a case in point for how contemporary web solutions can be used to build a single integrated educational system that facilitates teaching and learning in academic institutions.

2. Related work / Project

2.1. E-Learning Management Systems (LMS)

2.1.1. Moodle

Moodle is one of the most widely adopted open source LMS platforms. It offers course management, assessment tools, and collaborative features. While highly extensible through plugins, Moodle's complex interface and steep learning curve make it less accessible for smaller institutions in developing regions.(https://moodle.org)

2.1.2. Coursera and edX

Coursera and edX represent massive open online course (MOOC) platforms that democratize access to university-level education . These platforms leverage video lectures, peer assessments, and verified certificates. However, their content is predominantly Western centric with limited Bengali language support and payment options suitable for local markets. (https://www.coursera.org, https://www.edx.org)

2.1.3. Udemy

Udemy operates as a marketplace model where instructors independently create and sell courses. While this approach scales content creation, it lacks quality control and structured learning paths. (https://www.udemy.com)

2.1.4. 10 Minute School

10 Minute School is Bangladesh's leading edtech startup which offers live classes, recorded lectures, and exam preparation materials primarily for school and university admission tests. The downside is it focuses less on professional skill development and certificate courses. (https://10minuteschool.com)

2.1.5. Shikho

Shikho provides adaptive learning experiences for Bengali students using AI-driven personalization. Their mobile-first approach addresses connectivity challenges but lacks comprehensive instructor-led courses and project-based learning. (https://shikho.com)

2.2 Assessment and Evaluation Systems

2.2.1. Khan Academy

Khan Academy offers mastery-based learning with adaptive exercises and instant feedback mechanisms. Their approach to assessment enhances engagement but is primarily designed for K-12 mathematics and sciences. (https://www.khanacademy.org)

2.2.2. Gradescope

Gradescope offers AI-assisted grading for programming assignments and handwritten exams. While powerful for automated assessment, it requires institutional licenses and lacks integration with course delivery. (https://www.gradescope.com)

2.3. Content Delivery and Video Hosting

2.3.1. YouTube

YouTube serves as the de facto platform for educational video content, leveraging CDN infrastructure for global reach. However, it lacks structured curriculum organization, assessment tools, and progress tracking. (https://www.youtube.com)

2.3.2. Vimeo

Vimeo offers higher quality video hosting with privacy controls suitable for paid courses. Many educators use Vimeo embedded in custom LMS solutions.(https://vimeo.com)

2.4. Personalization and Adaptive Learning

2.4.1. Duolingo

Duolingo demonstrates effective use of spaced repetition algorithms and adaptive difficulty adjustment in language learning . Their A/B testing culture continuously optimizes learning pathways. While highly effective for vocabulary acquisition, this approach is less applicable to complex professional skills. (https://www.duolingo.com)

2.4.2. Knewton (now part of Wiley)

Knewton developed sophisticated adaptive learning engines that adjust content recommendations based on learner performance patterns . However, such systems require large datasets and sophisticated machine learning infrastructure.
(https://www.wiley.com/en-us/grow/teach-learn/teacher-resources/courseware/knewton-alta/)

2.5. Collaboration and Community Features

2.5.1. Stack Overflow

Stack Overflow exemplifies effective Q and A architecture with voting, reputation systems, and content moderation. Educational platforms increasingly adopt similar mechanisms for peer support.(https://stackoverflow.com)

2.6. Mobile Learning Applications

2.6.1. SoloLearn

SoloLearn offers bite-sized coding lessons optimized for mobile devices. It has peer code review features. While effective for introductory programming, it lacks depth for professional certification. (https://www.sololearn.com)

2.7. Research Gap

Existing e-learning platforms either target global markets without local customization (Coursera, Udemy) or focus on K-12 exam preparation within Bangladesh (10 Minute School, Shikho). A gap exists for professional skill development platforms with:

- ➔ Bengali language content
- ➔ Local payment gateway integration
- ➔ Hybrid assessment mechanisms (automated MCQ + manual project review)
- ➔ Instructor revenue sharing models

প্রজ্ঞা addresses this gap by providing a comprehensive learning management system. It is tailored to Bangladesh's professional education market, combining best practices from global platforms with localization requirements essential for adoption in emerging markets.

3. Design

3.1 Use Case Diagram

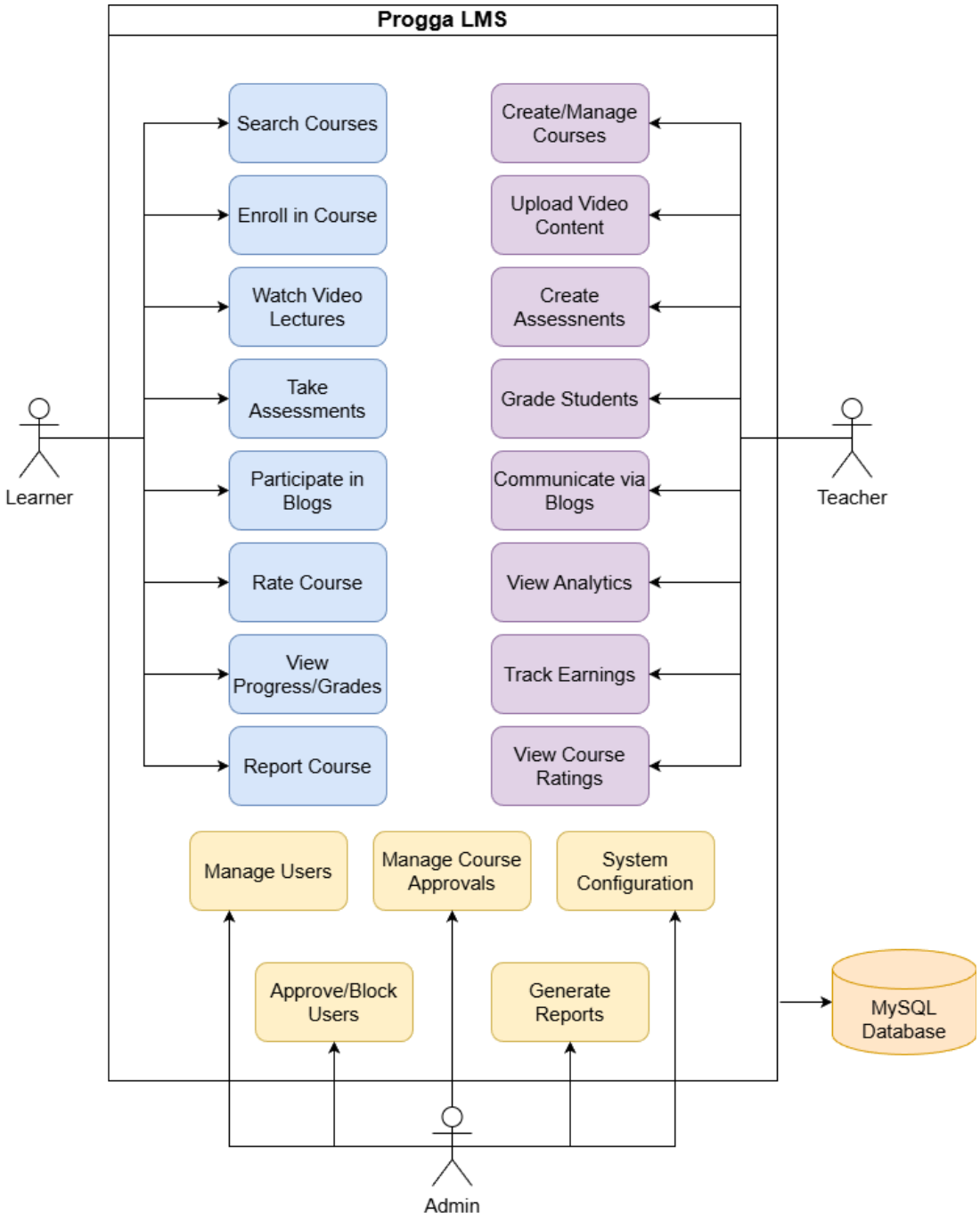


Figure 1: Level 0 Use Case Diagram of Progga LMS

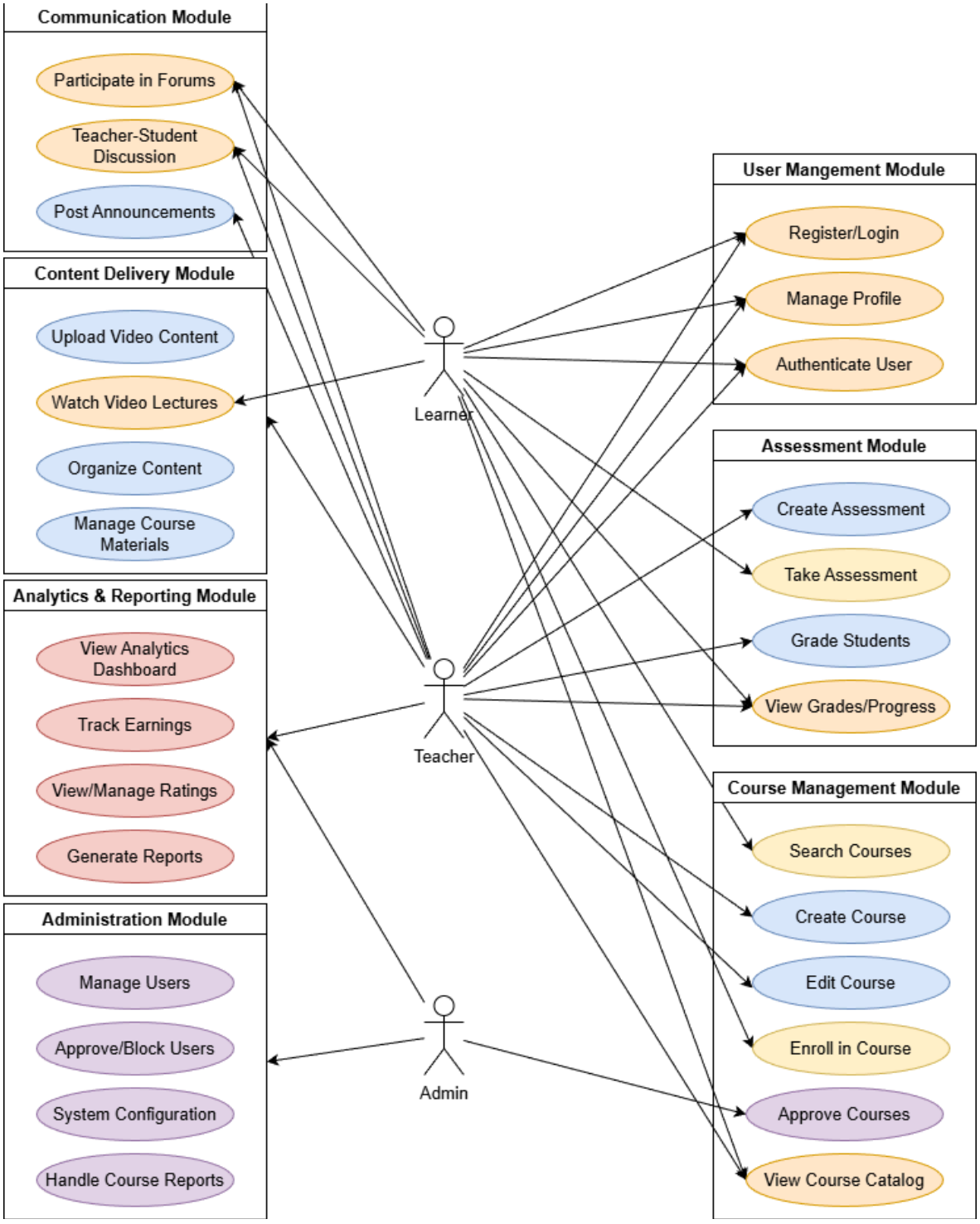
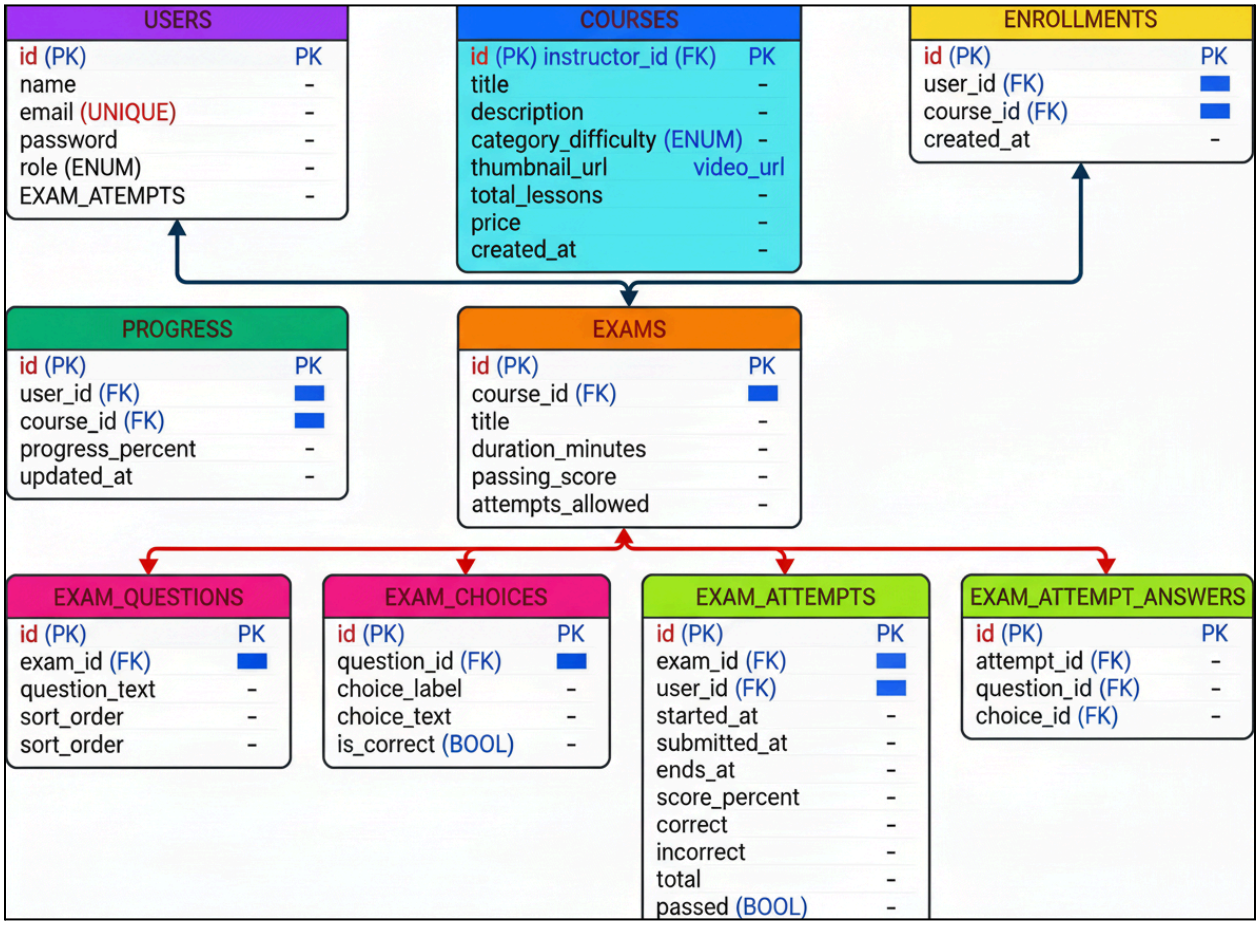


Figure 2: Level 1 Use Case Diagram of Progga LMS

3.2 Database Schema



4. User manual (How to install and use the system)

Step 1: Download and Install XAMPP

- ➔ Download XAMPP from https://www.apachefriends.org/
- ➔ Run the installer (xampp-windows-x64-8.2.4-0-VS16-installer.exe)
- ➔ Select components:
  - ◆ Apache
  - ◆ MySQL
  - ◆ PHP
  - ◆ phpMyAdmin
- ➔ Choose installation directory (default: C:\xampp)
- ➔ Click "Next" and complete installation

Step 2: Start XAMPP Services

- ➔ Launch XAMPP Control Panel
- ➔ Click "Start" for Apache
- ➔ Click "Start" for MySQL
- ➔ Verify services are running (green highlights)

Step 3: Create Database

- ➔ Open browser and navigate to http://localhost/phpmyadmin
- ➔ Click "New" in the left sidebar
- ➔ Enter database name: pragya_db
- ➔ Select collation: utf8mb4_unicode_ci
- ➔ Click "Create"

Step 4: Import Database Schema

- ➔ Click on pragya database
- ➔ Click "Import" tab
- ➔ Click "Choose File" and select database/schema.sql
- ➔ Scroll down and click "Go"
- ➔ Wait for "Import has been successfully finished" message

Database Schema file: https://jumpshare.com/share/59cC8CdvtC0KsuJLHPxg

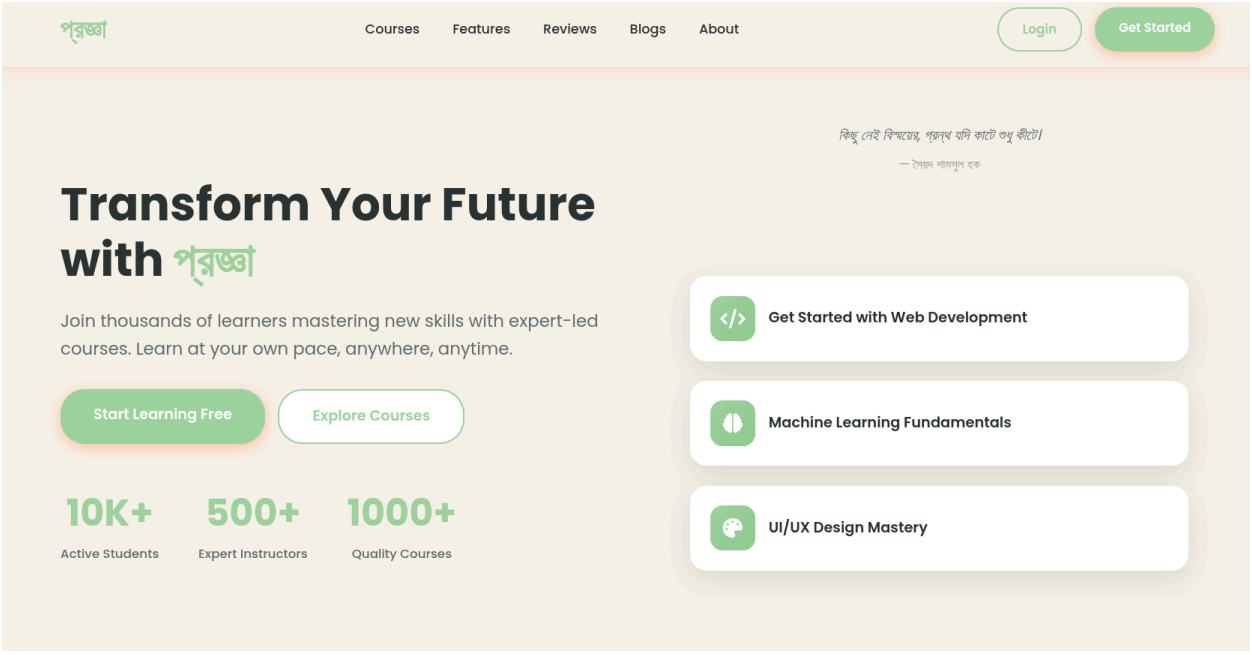
Step 5: Test Installation

- ➔ Open browser: http://localhost/proj/index.php and this will open the প্রজ্ঞা homepage

5. Conclusion

The project প্রজ্ঞা is a complete online learning system that lets students learn from courses, watch videos, and take MCQ-based exams in one place. It also gives teachers tools to create courses and exams, and admins tools to manage users, courses, and the overall platform. This shows that a practical, locally focused e-learning platform can be built with standard web technologies and can be extended later with features like better analytics, smarter exam logic, and local payment integration.

6. Screenshots of the Website



All Subjects

All Difficulties



CSS Essentials

Students in “Modern CSS Essentials: From Selectors to Layouts” will learn to: 1...

Beginner • 10 lessons



2

2...

Beginner • 10 lessons



Teaching cyber-bullying

Learn Cyber-bullying techniques <3...

Advanced • 1 lessons



1234

ml course...

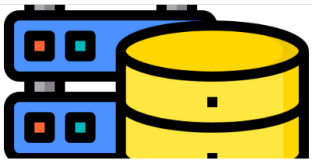
Intermediate • 10 lessons

পূরণ্ডা Home Courses My Courses My Dashboard My Blogs Blogs Discount T Logout

My Enrolled Courses

Track your learning progress and continue your education journey

2	0	0	0%
Enrolled Courses	Completed	Lessons Completed	Average Progress



Teaching cyber-bullying

1 Lessons Advanced

Progress 0%

0 of 1 lessons completed

Start Learning Take Exam



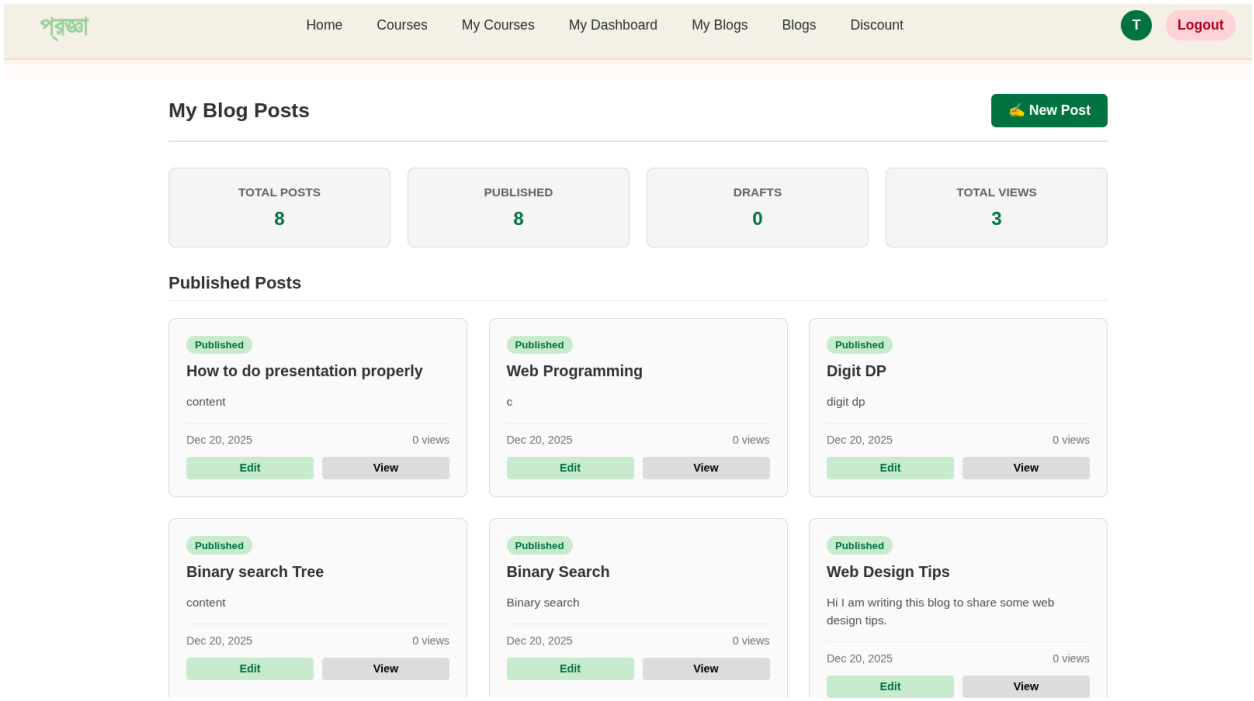
Get Started with Html

5 Lessons Beginner

Progress 0%

0 of 5 lessons completed

Start Learning Take Exam


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My Courses
My Dashboard
My Blogs
Blogs
Discount
T
Logout
My Blog Posts
✍ New Post
TOTAL POSTS
8
PUBLISHED
8
DRAFTS
0
TOTAL VIEWS
3
Published Posts
Published
How to do presentation properly
content
Dec 20, 2025
0 views
Edit
View
Published
Web Programming
c
Dec 20, 2025
0 views
Edit
View
Published
Digit DP
digit dp
Dec 20, 2025
0 views
Edit
View
Published
Binary search Tree
content
Dec 20, 2025
0 views
Edit
View
Published
Binary Search
Binary search
Dec 20, 2025
0 views
Edit
View
Published
Web Design Tips
Hi I am writing this blog to share some web design tips.
Dec 20, 2025
0 views
Edit
View

CSS Essentials

By teacher • 10 Lessons • Beginner

Students in “Modern CSS Essentials: From Selectors to Layouts” will learn to: 1. Understand how CSS works with HTML: selectors, the cascade, specificity, and inheritance. 2. Style text and colors using properties like color, font-family, font-size, line-height, and more. 3. Work with the box model (margin, padding, border, width/height) to control spacing and layout.4. Use modern layout systems such as Flexbox (for one-dimensional layouts) and basic Grid concepts for multi-column sections.5. Apply responsive design basics with media queries so pages look good on mobile, tablet, and desktop

What You'll Learn

- No learning outcomes added yet.

Student Reviews

5.0 ★ 1 ratings • 1 reviews

★★★★★

Student ★★★★★

Joss

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CSS Essentials FREE



Students in “Modern CSS Essentials: From Selectors to Layouts” will learn to: 1...

Price: Free

1234 FREE



ml course...

Price: Free

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Welcome back, al habibi!

Continue your learning journey

Enrolled Courses **3**

Completed Courses **0**

In Progress **3**

My Courses



Teaching cyber-bullying

English

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Final Exam

Difficulty: EASY

Total Questions: 2

Time Limit: 30 minutes

Passing Score: 0

Attempts Allowed: 10

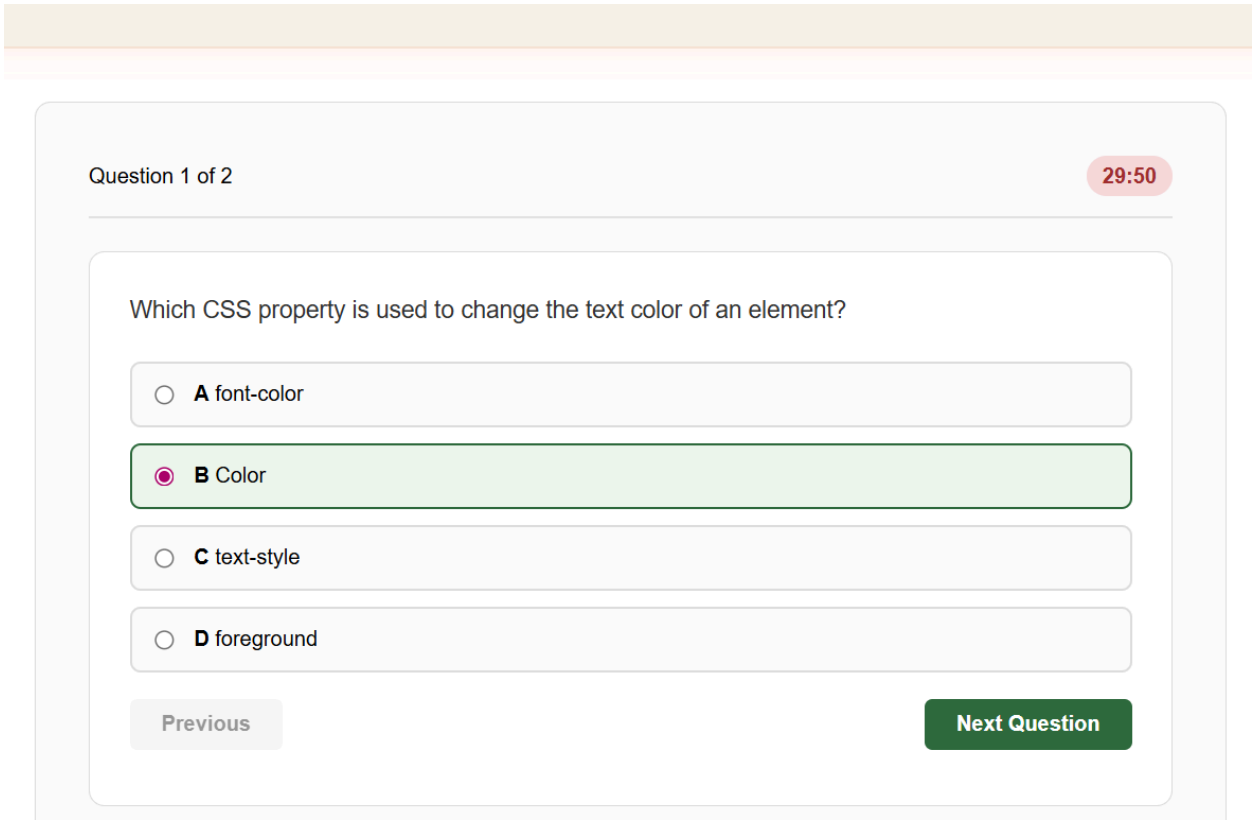
Attempts Used: 0

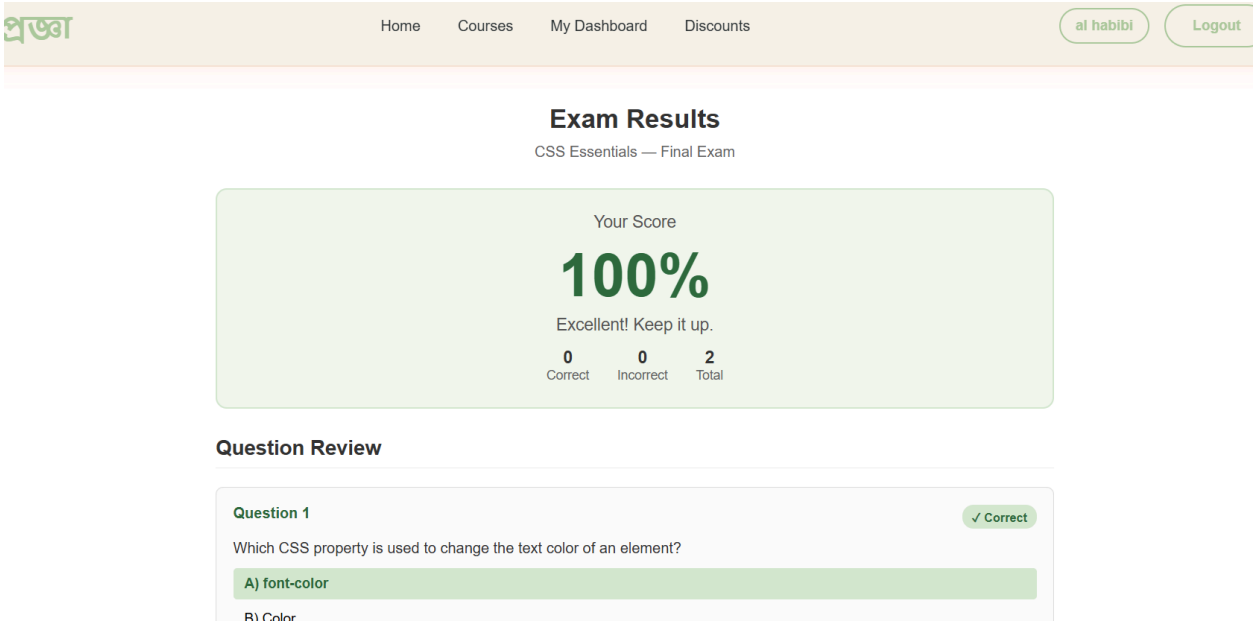
Exam Instructions

- Read each question carefully before answering.
- You can navigate between questions using "Previous" and "Next".
- Your answers are saved automatically.
- Good luck!

Start Exam

← Back to Course


Question 1 of 2
29:50
Which CSS property is used to change the text color of an element?
A font-color
B Color
C text-style
D foreground
Previous
Next Question


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Home
Courses
My Dashboard
Discounts
al habibi
Logout
Exam Results
CSS Essentials — Final Exam
Your Score
100%
Excellent! Keep it up.
0
Correct
0
Incorrect
2
Total
Question Review
Question 1
✓ Correct
Which CSS property is used to change the text color of an element?
A) font-color
B) Color